



22PH202 – Electromagnetism and Modern Physics

Unit II – Magnetism

2.6 Electromagnetic induction

Whenever the magnetic flux linked with a closed coil changes, an emf (electromotive force) is induced and hence an electric current flows in the circuit. This current is called an induced current and the emf giving rise to such current is called an induced emf. This phenomenon is known as electromagnetic induction.

2.6.1 Faraday’s first experiment

In this experiment, Faraday connected a coil to a galvanometer, as shown in the figure above. A bar magnet was pushed towards the coil, such that the north pole is pointing towards the coil. As the bar magnet is shifted, the pointer in the galvanometer gets deflected, thus indicating the presence of current in the coil under consideration. It is observed that when the bar magnet is stationary, the pointer shows no deflection and the motion lasts only till the magnet is in motion. Here, the direction of the deflection of the pointer depends upon the direction of motion of the bar magnet. Also, when the south pole of the bar magnet is moved towards or away from the coil, the deflections in the galvanometer are opposite to that observed with the north-pole for similar movements. Apart from this, the deflection of the pointer is larger or smaller depending upon the speed with which it is pulled towards or away from the coil. The same effect is observed when instead of the bar magnet, the coil is moved and the magnet is held stationary. This shows that only the relative motion between the magnet and the coil are responsible for the generation of current in the coil.

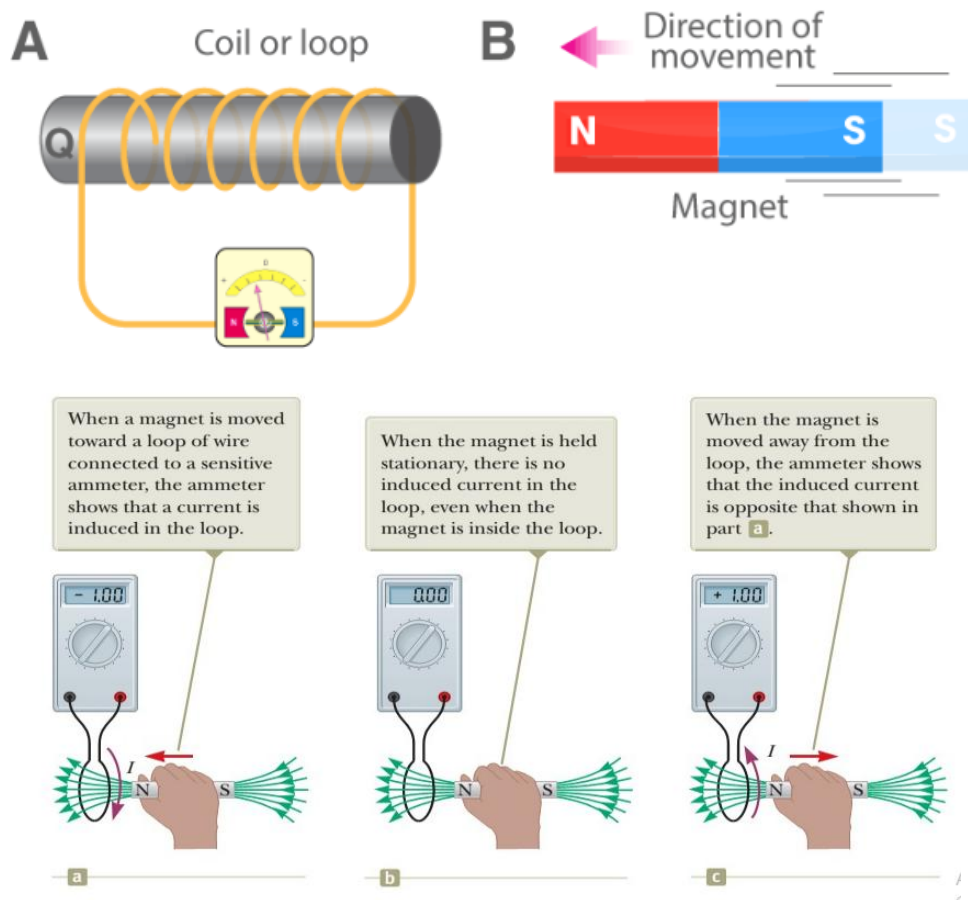


Figure 16. Faraday's first experiment

2.6.2 Faraday's second experiment

In the second experiment, Faraday replaced the bar magnet by a second current-carrying coil that was connected to a battery. Here, the current in the coil due to the connected battery produced a steady magnetic field, which made the system analogous to the previous one. As we move the second coil towards the primary coil, the pointer in the galvanometer undergoes deflection, which indicates the presence of the electric current in the first coil. Similar to the above case, here too, the direction of the deflection of the pointer depends upon the direction of motion of the secondary coil towards or away from the primary coil. Also, the magnitude of deflection depends upon the speed with which the coil is moved. All these results show that the system in the second case is analogous to the system in the first experiment.

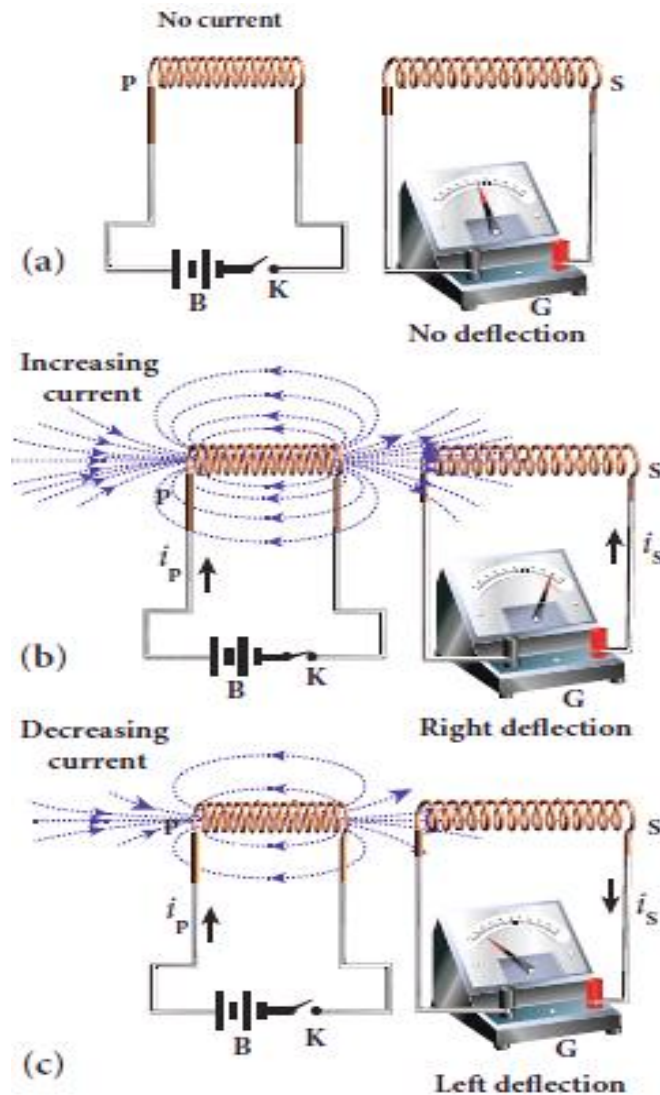


Figure 17. Faraday's second experiment

2.7 Lenz's Law

Lenz's law states that the direction of the induced current is such that it always opposes the cause responsible for its production. Faraday discovered that whenever magnetic flux linked with a coil changes, an electric current is induced in the circuit. Here the flux change is the cause while the induction of current is the effect. Lenz's law says that the effect always opposes the cause. Therefore, the induced current would flow in a direction so that it could oppose the flux change.

Lenz's law applications are plenty. Some of them are listed below-

- Eddy current balances
- Metal detectors

- Eddy current dynamometers
- Braking systems on train
- AC generators
- Card readers
- Microphones

The truth of Lenz's law can be established on the basis of the law of **conservation of energy**. The explanation is as follows: According to Lenz's law, when a magnet is moved either towards or away from a coil, the induced current produced opposes its motion. As a result, there will always be a resisting force on the moving magnet. Work has to be done by some external agency to move the magnet against this resisting force. Here the mechanical energy of the moving magnet is converted into the electrical energy which in turn, gets converted into Joule heat in the coil i.e., energy is converted from one form to another.

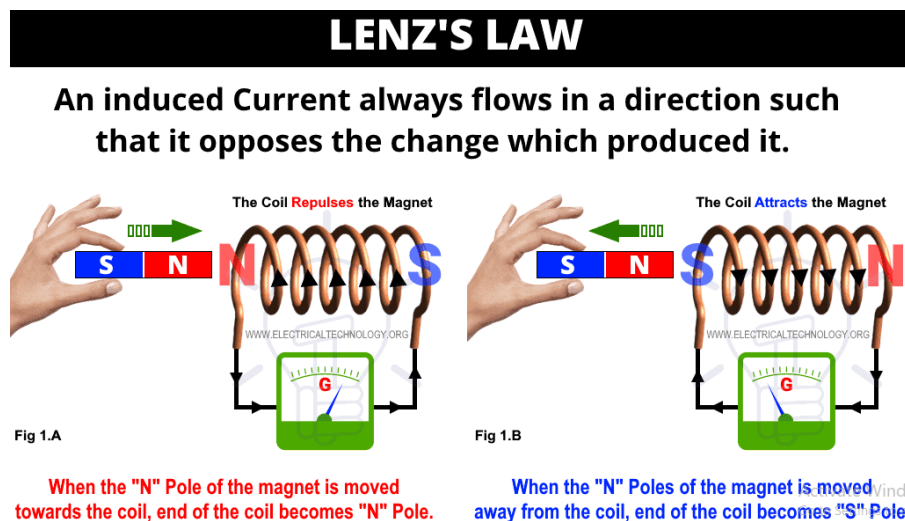


Figure 18. Lenz's law experiment

2.8 Fleming's hand rule

When a conductor moves in a magnetic field, the direction of motion of the conductor, the field and the induced current are given by Fleming's right hand rule.

Fleming's Right Hand Rule states that if we arrange our thumb, forefinger and middle finger of the right-hand perpendicular to each other, then the thumb points towards the direction of the motion of the conductor relative to the magnetic field, the forefinger points towards the direction of the magnetic field and the middle finger points towards the direction of the induced current.

FLEMING'S RIGHT HAND RULE

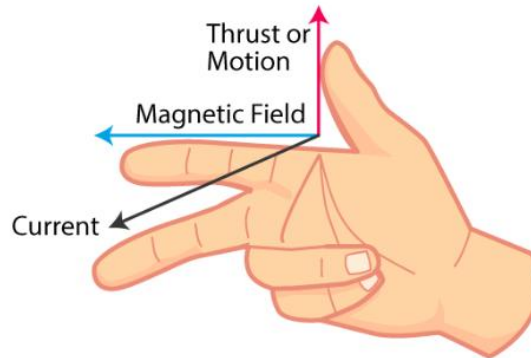


Figure 19. Fleming's right hand rule

Fleming's left hand rule

When a current-carrying conductor is placed in an external magnetic field, the conductor experiences a force perpendicular to both the field and the current flow's direction. Fleming's left-hand rule is used to find the direction of the force acting on the current carrying conductor placed in a magnetic field.

FLEMING'S LEFT HAND RULE

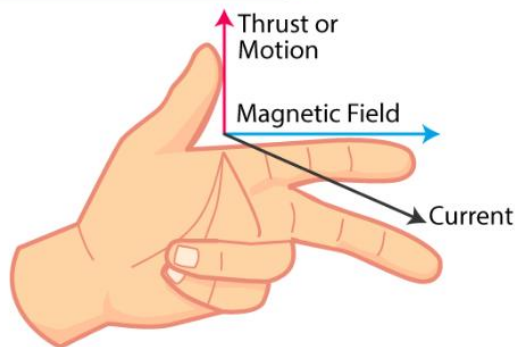


Figure 20. Fleming's left hand rule

2.9 Magnetic field by current carrying conductors

Oersted experiment

In 1820 Hans Christian Oersted, while preparing for his lecture in physics, noticed that electric current passing through a wire deflects the nearby magnetic needle in the compass. By proper investigation, he observed that the deflection of magnetic needle is due to the change in magnetic field produced around current carrying conductor. When the direction of current is reversed, the magnetic needle is

deflected in the opposite direction. This led to the development of the theory ‘electromagnetism’ which unifies two branches in physics namely, electricity and magnetism.

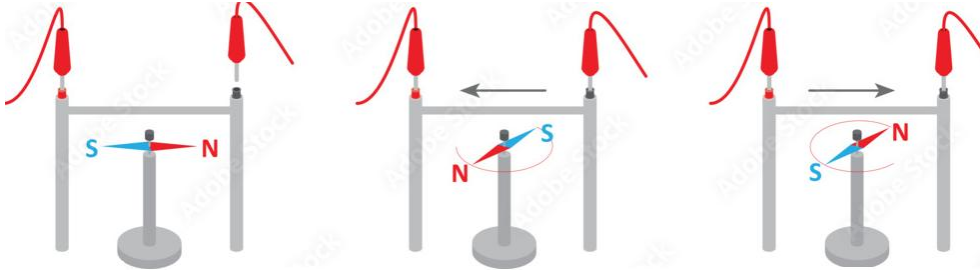


Figure 21. Oersted's experiment

Current carrying straight conductor

Suppose we keep a magnetic compass near a current-carrying straight conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. Tracing out the direction shown by magnetic needle, we can draw the magnetic field lines at a distance. For a straight current-carrying conductor, the nature of magnetic field is like concentric circles having their common centre on the axis of the conductor.

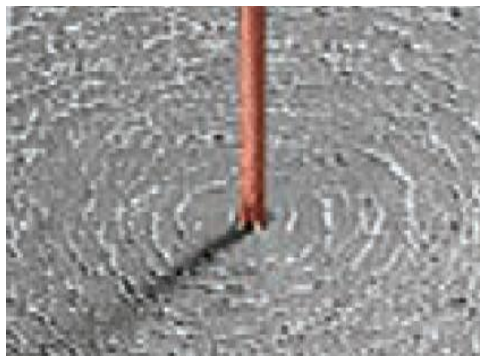


Figure 22. Magnetic field of current carrying straight conductor

Circular coil carrying current

Suppose we keep a magnetic compass near a current-carrying circular conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. We can notice that at the points A and B in the vicinity of the coil, the magnetic field lines are circular. The magnetic field lines are nearly parallel to each other near the centre of the loop, indicating that the field present near the centre of the coil is almost uniform.

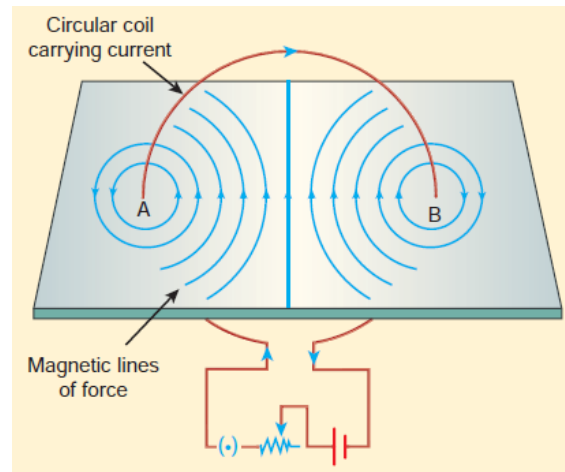


Figure 23. Magnetic field of current carrying circular coil

2.10 Maxwell's right hand cork screw rule

This rule can also be used to find the direction of the magnetic field around the current-carrying conductor. If we rotate a right-handed screw using a screw driver, then the direction of current is same as the direction in which screw advances and the direction of rotation of the screw gives the direction of the magnetic field.

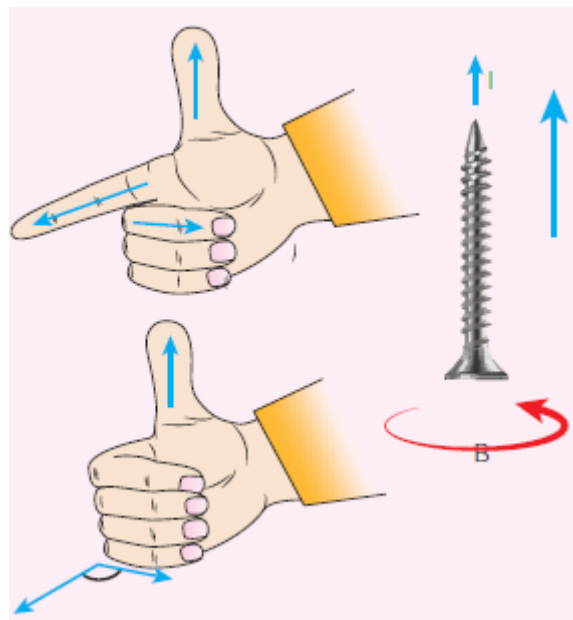


Figure 24. Maxwell's right hand cork screw rule

2.11 Biot-Savart law

Biot-Savart's law is an equation that gives the magnetic field produced due to a current carrying segment. This segment is taken as a vector quantity known as the current element.

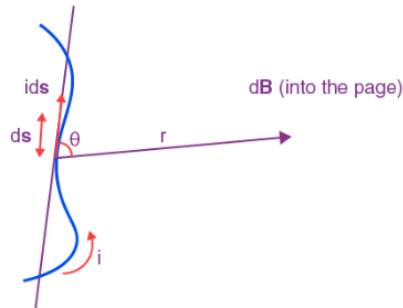


Figure 25. Biot-Savart law

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

- The law is also valid for a current consisting of charges flowing through space
- **ds** represent the length of a small segment of space in which the charges flow
- Example: electron beam in a TV set

2.12 Magnetic force between two parallel conductor

If the currents are in the:

- same direction the wires attract each other.
- opposite directions the wires repel each other.

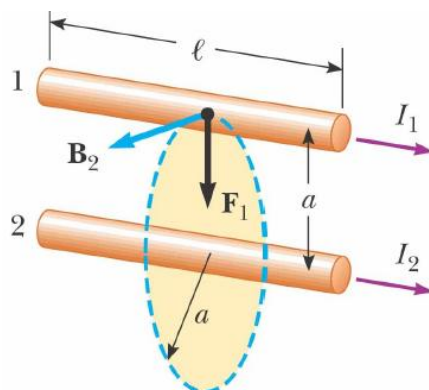


Figure 26. Current carrying parallel conductors

2.13 Ampère's circuital law

Ampère's law: The line integral of magnetic field over a closed loop is μ_0 times net current enclosed by the loop.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

where I_{enclosed} is the net current linked by the closed loop C. Note that the line integral does not depend on the shape of the path or the position of the conductor with the magnetic field.

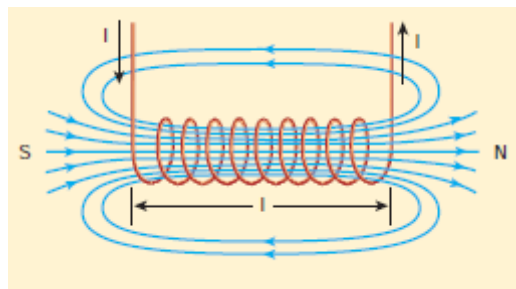


Figure 27. Magnetic field line of solenoid

2.14 Inductors

An Inductor is a passive electrical component consisting of a coil of wire which is designed to take advantage of the relationship between magnetism and electricity as a result of an electric current passing through the coil. Inductor is a device used to store energy in a magnetic field when an electric current flows through it.

Inductance is the tendency of an electrical conductor to oppose a change in the electric current flowing through it. The electric current produces a magnetic field around the conductor. The magnetic field strength depends on the magnitude of the electric current, and follows any changes in the magnitude of the current.

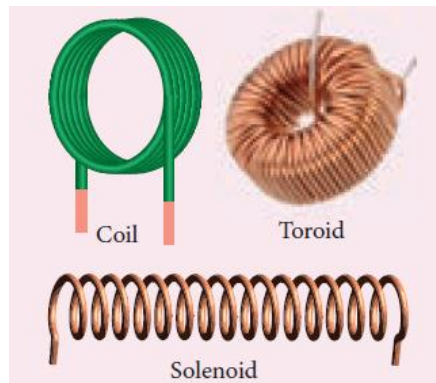


Figure 28. Examples of inductor

Depending on the type of material used inductors can be classified as follows:

- Iron Core Inductor
- Air Core Inductor
- Iron Powder Inductor

Ferrite Core Inductor which is divided into,

- ❖ Soft Ferrite
- ❖ Hard Ferrite

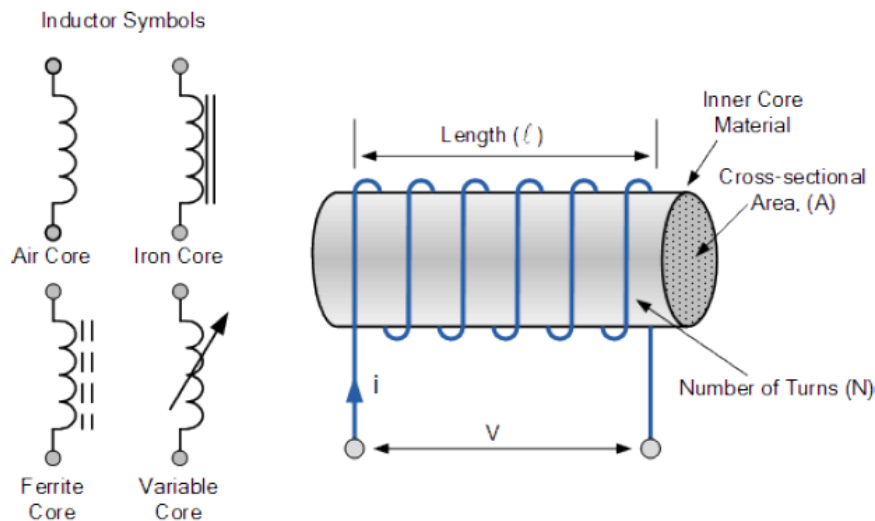


Figure 29. Symbol of inductors

Self-Inductance

An electric current flowing through a coil will set up a magnetic field around it. Therefore, the magnetic flux of the magnetic field is linked with that coil itself. If this flux is changed by changing

the current, an emf is induced in that same coil (Figure 4.18). This phenomenon is known as self-induction. The emf induced is called self-induced emf.

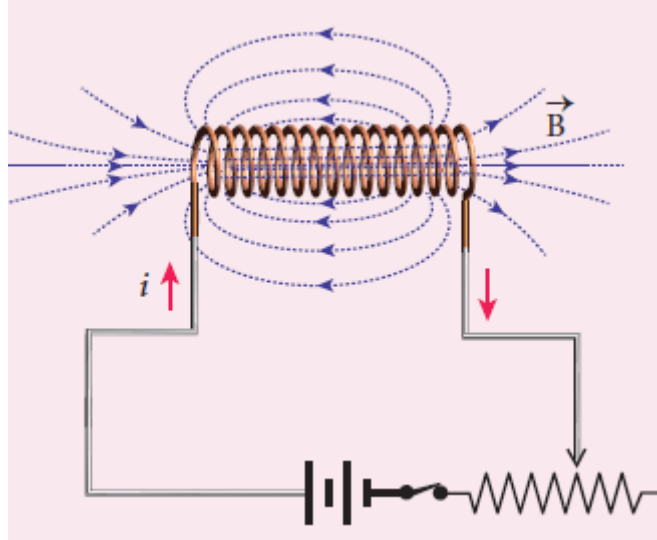


Figure 30. Self-inductance

Mutual Inductance

When an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil. This phenomenon is known as mutual induction and the emf induced is called mutually induced emf.

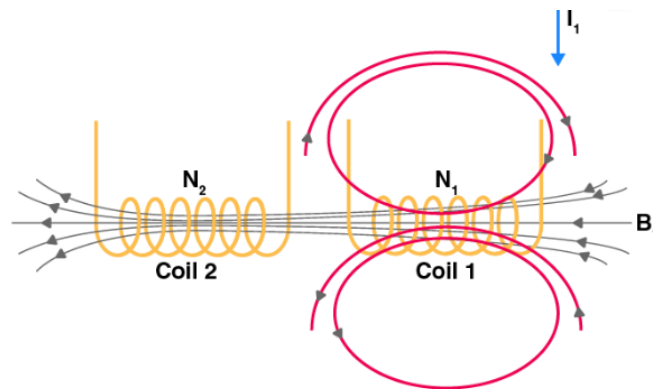


Figure 31. Mutual-inductance

2.15 Inductors in series and parallel

Inductors in series are the process of adding inductors in a sequence. In an electric circuit, an inductor is an electrical component that can be visualized as electrical inductance to store energy in its

magnetic field. Power supplies, Transformers, Radios, RADARs, etc., are some applications of inductors or groups of inductors in series.

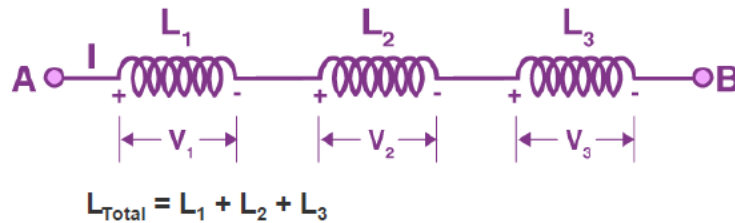


Figure 32. Inductors in series

In the connection of **Inductors in Parallel**, the voltage across each inductor will be the same, and the current divided across the inductors. The least opposing inductor will have the maximum current through it. The reciprocal of equivalent inductance is the sum of the reciprocal of individual inductance present in the circuit.

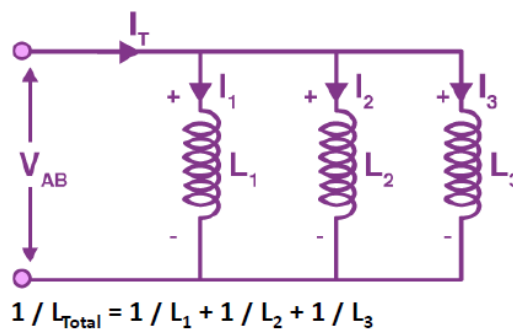


Figure 33. Inductors in parallel

2.16 Eddy current

According to Faraday's law of electromagnetic induction, an emf is induced in a conductor when the magnetic flux passing through it changes. However, the conductor need not be in the form of a wire or coil. Even for a conductor in the form of a sheet or plate, an emf is induced when magnetic flux linked with it changes. But the difference is that there is no definite loop or path for induced current to flow away. As a result, the induced currents flow in concentric circular paths. As these electric currents resemble eddies of water, these are known as Eddy currents. They are also called Foucault currents.

- ❖ When eddy currents flow in the conductor, a large amount of energy is dissipated in the form of heat. The energy loss due to the flow of eddy current is inevitable but it can be reduced to a greater extent with suitable measures

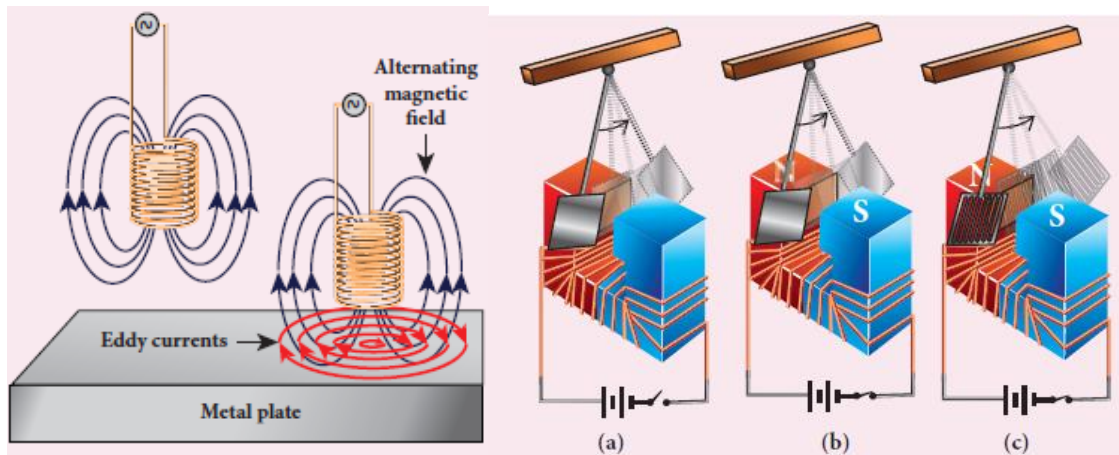


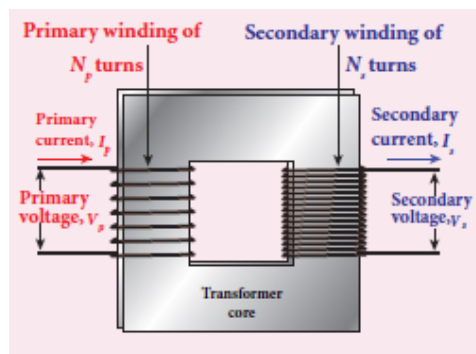
Figure 34. Eddy current

Applications of eddy current

- Induction stove
- Eddy current brake
- Eddy current testing
- Electromagnetic damping

2.17 Transformer

Transformer is a stationary device used to transform electrical power from one circuit to another without changing its frequency. The applied alternating voltage is either increased or decreased with corresponding decrease or increase of current in the circuit. If the transformer converts an alternating current with low voltage into an alternating current with high voltage, it is called step-up transformer. On the contrary, if the transformer converts alternating current with high voltage into an alternating current with low voltage, then it is called step-down transformer.



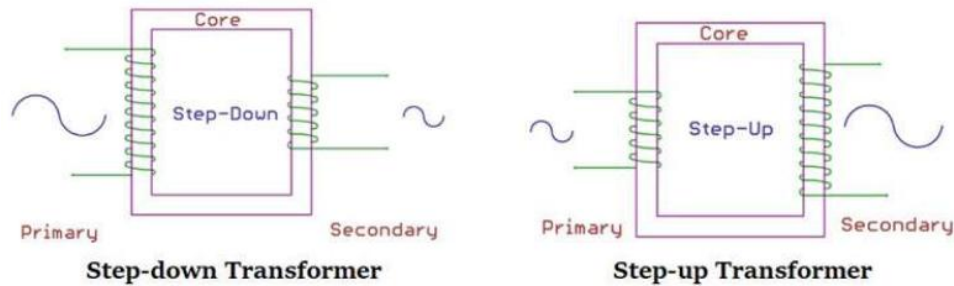


Figure 35. Transformer

Working principle

The principle of transformer is the mutual induction between two coils. That is, when an electric current passing through a coil changes with time, an emf is induced in the neighbouring coil.

Transformer works on the principle of mutual induction of two coils or Faraday Law's of Electromagnetic Induction. When current in the primary coil is changed the flux linked to the secondary coil also changes. Consequently an EMF is induced in the secondary coil due to Faraday law's of electromagnetic induction.

The transformer is based on two principles: first, that an electric current can produce a magnetic field, and second, that a changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). Changing the current in the primary coil changes the magnetic flux that is developed. The changing magnetic flux induces a voltage in the secondary coil.

Energy loss

Transformers do not have any moving parts so that its efficiency is much higher than that of rotating machines like generators and motors. But there are many factors which lead to energy loss in a transformer.

- ❖ **Core loss or Iron loss**
- ❖ **Copper loss**
- ❖ **Flux leakage**

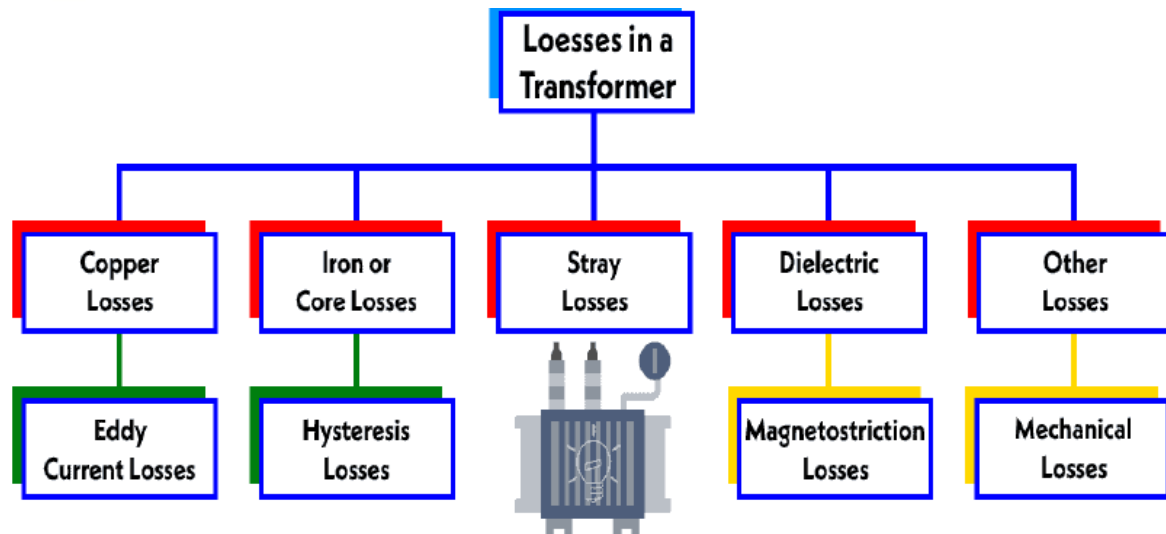


Figure 36. Losses in transformers

Limitation of transformer:

A transformer is an AC machine that steps up or steps down an alternating voltage or current. A transformer being an AC machine however cannot step up or down a DC voltage or DC current.

2.18 Comparison of magnetic and electric field

Comparison of Magnetic to Electric Field

Magnetic Field	Electric Field
• B proportional to r^2 • Vector • Perpendicular to $\mathbf{F}_B$, $d\mathbf{s}$, $\mathbf{r}$ • Magnetic field lines have no beginning and no end; they form continuous circles • Biot-Savart Law • Ampere's Law (where there is symmetry)	• E proportional to r^2 • Vector • Same direction as $\mathbf{F}_E$ • Electric field lines begin on positive charges and end on negative charges • Coulomb's Law • Gauss's Law (where there is symmetry)